

An Information-Theoretic Perspective on Covert Uplink Communication in FD-ISAC Systems

System Model. We consider a simplified single-antenna full-duplex (FD) system that isolates the fundamental interaction between an uplink covert transmission and a concurrent downlink waveform. Alice communicates covertly with an FD base station (BS), while a passive warden Willie attempts to determine whether Alice is active. During a block of n channel uses, the BS simultaneously transmits an independent Gaussian downlink waveform

$$S_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, P_d), \quad i = 1, \dots, n, \quad (1)$$

where P_d is fixed. All channel coefficients are assumed constant over the block and known. Willie knows the distribution and power of S^n , but not its realization.

After self-interference cancellation, the BS observes

$$Y_i = h_{ab}X_i + \tilde{N}_{b,i}, \quad \tilde{N}_{b,i} \sim \mathcal{CN}(0, N_b(P_d)), \quad (2)$$

where $N_b(P_d) \triangleq \sigma_b^2 + \beta|h_{\text{SI}}|^2 P_d$ models receiver noise and residual self-interference. Here h_{ab} is the Alice–BS channel, h_{SI} is the self-interference channel, and $\beta \geq 0$ denotes the residual self-interference factor.

Willie’s observation is

$$Z_i = g_{aw}X_i + g_{bw}S_i + N_{w,i}, \quad N_{w,i} \sim \mathcal{CN}(0, \sigma_w^2), \quad (3)$$

where g_{aw} and g_{bw} denote the Alice–Willie and BS–Willie channels, respectively. Defining

$$V_i \triangleq g_{bw}S_i + N_{w,i} \sim \mathcal{CN}(0, N_w(P_d)), \quad N_w(P_d) \triangleq \sigma_w^2 + |g_{bw}|^2 P_d, \quad (4)$$

the surveillance channel becomes $Z_i = g_{aw}X_i + V_i$.

Code Definition. For blocklength n , Alice employs a fixed (n, M_n) codebook

$$\mathcal{C}_n = \{x^n(1), \dots, x^n(M_n)\} \subset \mathbb{C}^n, \quad (5)$$

with a uniformly distributed message $M \sim \text{Unif}\{1, \dots, M_n\}$. The encoder maps $M = m$ to the deterministic codeword $x^n(m)$, subject to the per-codeword power constraint

$$\frac{1}{n} \|x^n(m)\|^2 \leq P_a, \quad \forall m. \quad (6)$$

The BS decoder $\phi_n : \mathbb{C}^n \rightarrow \{1, \dots, M_n\}$ produces $\hat{M} = \phi_n(Y^n)$, and reliability requires

$$P_e^{(n)} \triangleq \Pr(\hat{M} \neq M) \rightarrow 0. \quad (7)$$

Willie’s Hypothesis Test and Coverttness. Under Alice’s silence,

$$\mathcal{H}_0 : \quad Z^n = V^n, \quad P_0^{(n)} = \mathcal{CN}(\mathbf{0}, N_w(P_d)\mathbf{I}_n). \quad (8)$$

Under $\mathcal{H}_1$, conditioned on message $M = m$ and the fixed codebook,

$$Z^n = g_{aw}x^n(m) + V^n, \quad (9)$$

and therefore

$$P_{Z^n|M=m, \mathcal{H}_1, \mathcal{C}_n} = \mathcal{CN}(g_{aw}x^n(m), N_w(P_d)\mathbf{I}_n). \quad (10)$$

Since Willie does not know M , the actual distribution induced by the fixed codebook is the Gaussian mixture

$$P_{1, \mathcal{C}_n}^{(n)} = \frac{1}{M_n} \sum_{m=1}^{M_n} \mathcal{CN}(g_{aw}x^n(m), N_w(P_d)\mathbf{I}_n). \quad (11)$$

We impose the coverttness constraint

$$D(P_{1, \mathcal{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta, \quad (12)$$

for a prescribed $\delta > 0$. The fundamental objective is to characterize the maximum codebook size

$$M_n^*(\delta) = \max_{\mathcal{C}_n, \phi_n} \left\{ M_n : P_e^{(n)} \rightarrow 0, D(P_{1, \mathcal{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta \right\}, \quad (13)$$

and, in particular, the asymptotic growth of $\log M_n^*(\delta)$ as $n \rightarrow \infty$ and its dependence on the concurrent downlink power P_d .

Theorem 1 (Achievable Square-Root Law for FD-ISAC). Assume $g_{aw} \neq 0$, $P_a > 0$ is fixed, and

$$\frac{|h_{ab}|^2}{N_b(P_d)} > \frac{|g_{aw}|^2}{N_w(P_d)}. \quad (14)$$

For every prescribed $\delta > 0$, there exists a sequence of deterministic fixed codebooks satisfying the per-codeword power constraint, vanishing decoding error probability, and

$$D\left(P_{1, \mathcal{C}_n}^{(n)} \parallel P_0^{(n)}\right) \leq \delta. \quad (15)$$

Moreover, the maximum admissible codebook size satisfies

$$\liminf_{n \rightarrow \infty} \frac{\log M_n^*(\delta)}{\sqrt{n}} \geq \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}. \quad (16)$$

Thus the achievable covert information grows on the square-root scale, $\log M_n = \Theta(\sqrt{n})$.

Physical Interpretation. The achievable square-root coefficient depends on the concurrent downlink power P_d through

$$\frac{N_w(P_d)}{N_b(P_d)} = \frac{\sigma_w^2 + |g_{bw}|^2 P_d}{\sigma_b^2 + \beta |h_{SI}|^2 P_d}. \quad (17)$$

Increasing P_d therefore has two competing effects: it increases the uncertainty at Willie through the BS–Willie link, while simultaneously increasing the residual self-interference at the BS. In particular, the achievable coefficient increases with P_d if

$$|g_{bw}|^2 \sigma_b^2 > \beta |h_{SI}|^2 \sigma_w^2, \quad (18)$$

decreases with P_d when the inequality is reversed, and is independent of P_d at the square-root order when equality holds. The downlink power also affects the feasibility condition in Theorem 1 through the relative effective qualities of the Alice–BS and Alice–Willie channels.

Proof. The proof first identifies the covert signaling scale and derives the complementary reliability and resolvability requirements on the codebook size. It then verifies power admissibility, converts resolvability into the actual covertness criterion, and extracts a deterministic codebook sequence.

Random-Coding Construction. We use Gaussian random coding so that the ensemble-induced output law at Willie is analytically tractable, while the final deterministic codebook will be selected only after all constraints have been established. For each blocklength n , let $Q_n > 0$ and $M_n \in \mathbb{N}$. Generate the random codebook

$$\mathbf{C}_n = \{X^n(1), \dots, X^n(M_n)\}, \quad X_i(m) \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, Q_n), \quad (19)$$

independently over $i = 1, \dots, n$ and $m = 1, \dots, M_n$. A realization of $\mathbf{C}_n$ is denoted by

$$\mathcal{C}_n = \{x^n(1), \dots, x^n(M_n)\}.$$

For $M \sim \text{Unif}\{1, \dots, M_n\}$, Alice transmits $x^n(M)$. The BS employs the ML decoder

$$\phi_n(y^n) = \arg \min_{1 \leq m \leq M_n} \|y^n - h_{ab} x^n(m)\|^2. \quad (20)$$

To characterize the output distribution induced by the Gaussian codeword ensemble at Willie, recall that

$$Z_i = g_{aw} X_i + V_i, \quad V_i \sim \mathcal{CN}(0, N_w(P_d)).$$

Since X_i and V_i are independent Gaussian random variables,

$$Z_i \sim \mathcal{CN}(0, N_w(P_d) + |g_{aw}|^2 Q_n). \quad (21)$$

Independence across channel uses therefore gives the Gaussian reference distribution

$$\bar{P}_{1,n}^{(n)} \triangleq \mathcal{CN}(\mathbf{0}, [N_w(P_d) + |g_{aw}|^2 Q_n] \mathbf{I}_n). \quad (22)$$

Reference Covertiness and Power Scaling. This step determines the signaling-power scale for which the ensemble-induced Gaussian output can remain covert with respect to the innocent law. We first evaluate the divergence between the Gaussian reference distribution in (17) and the innocent distribution $P_0^{(n)}$ in (8). For zero-mean proper complex Gaussian distributions,

$$D(\mathcal{CN}(\mathbf{0}, \mathbf{\Sigma}_1) \parallel \mathcal{CN}(\mathbf{0}, \mathbf{\Sigma}_0)) = \log \frac{\det \mathbf{\Sigma}_0}{\det \mathbf{\Sigma}_1} + \text{tr}(\mathbf{\Sigma}_0^{-1} \mathbf{\Sigma}_1) - n.$$

Hence,

$$D(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}) = n \left[\frac{|g_{aw}|^2 Q_n}{N_w(P_d)} - \log \left(1 + \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} \right) \right]. \quad (23)$$

Imposing

$$D(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}) \leq \delta \quad (24)$$

gives

$$0 \leq \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} - \log \left(1 + \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} \right) \leq \frac{\delta}{n}. \quad (25)$$

Since $x - \log(1+x)$ is strictly increasing for $x \geq 0$ and vanishes only at $x = 0$, it follows that

$$Q_n \rightarrow 0. \quad (26)$$

The expansion

$$x - \log(1+x) = \frac{x^2}{2} + O(x^3), \quad x \rightarrow 0,$$

therefore yields

$$D(\bar{P}_{1,n}^{(n)} \| P_0^{(n)}) = \frac{n|g_{aw}|^4 Q_n^2}{2N_w^2(P_d)} (1 + o(1)). \quad (27)$$

Consequently,

$$Q_n \leq \frac{N_w(P_d)}{|g_{aw}|^2} \sqrt{\frac{2\delta}{n}} (1 + o(1)), \quad (28)$$

and hence

$$Q_n = O(n^{-1/2}). \quad (29)$$

Power-Sequence Selection. We now select a sequence on the covert power scale while retaining a strict divergence margin for the later finite-codebook approximation. Fix an arbitrary $0 < \varepsilon < 1$. Motivated by (23), choose

$$Q_n = (1 - \varepsilon) \frac{N_w(P_d)}{|g_{aw}|^2} \sqrt{\frac{2\delta}{n}}. \quad (30)$$

Substituting this choice into (22) gives

$$\begin{aligned} D(\bar{P}_{1,n}^{(n)} \| P_0^{(n)}) &= \frac{n|g_{aw}|^4 Q_n^2}{2N_w^2(P_d)} (1 + o(1)) \\ &= (1 - \varepsilon)^2 \delta (1 + o(1)) < \delta \end{aligned} \quad (31)$$

for all sufficiently large n . Moreover,

$$\sqrt{n} Q_n \rightarrow (1 - \varepsilon) \frac{N_w(P_d)}{|g_{aw}|^2} \sqrt{2\delta}, \quad (32)$$

and hence

$$Q_n = \Theta(n^{-1/2}). \quad (33)$$

Reliability at the BS. We next derive a sufficient upper-side condition on $\log M_n$ under which the BS can decode reliably. Under the Gaussian input $X_i \sim \mathcal{CN}(0, Q_n)$, Bob's channel in (2) induces

$$Y_i \sim \mathcal{CN}(0, N_b(P_d) + |h_{ab}|^2 Q_n).$$

Let $p_{Y|X}(\cdot|x)$ denote the conditional Gaussian density in (2), and let $p_{Y,n}$ denote the above output density. Define the single-letter information density by

$$i_{b,n}(x; y) \triangleq \log \frac{p_{Y|X}(y|x)}{p_{Y,n}(y)}, \quad (34)$$

and, by memorylessness,

$$i_{b,n}^{(n)}(X^n; Y^n) = \sum_{i=1}^n i_{b,n}(X_i; Y_i). \quad (35)$$

Its mean is the Gaussian-input mutual information

$$\begin{aligned} I_{b,n} &\triangleq \mathbb{E}[i_{b,n}(X_i; Y_i)] \\ &= \log \left(1 + \frac{|h_{ab}|^2 Q_n}{N_b(P_d)} \right), \end{aligned} \quad (36)$$

while direct evaluation of the Gaussian information density gives

$$\mathbb{E}[i_{b,n}^{(n)}(X^n; Y^n)] = nI_{b,n}, \quad \text{Var}[i_{b,n}^{(n)}(X^n; Y^n)] = \frac{2n|h_{ab}|^2 Q_n}{N_b(P_d) + |h_{ab}|^2 Q_n}. \quad (37)$$

Since the ML decoder minimizes the average decoding error for a fixed codebook, by Feinstein's random-coding bound, for any $\tau_n > 0$,

$$\mathbb{E}_{C_n}[P_e^{(n)}] \leq \Pr[i_{b,n}^{(n)}(X^n; Y^n) \leq \log M_n + \tau_n] + e^{-\tau_n}. \quad (38)$$

Define the reliability margin

$$\Delta_{b,n} \triangleq nI_{b,n} - \log M_n. \quad (39)$$

For $\Delta_{b,n} > 0$, choosing $\tau_n = \Delta_{b,n}/2$ and applying Chebyshev's inequality gives

$$\begin{aligned} \mathbb{E}_{C_n}[P_e^{(n)}] &\leq \frac{4 \text{Var}[i_{b,n}^{(n)}]}{\Delta_{b,n}^2} + e^{-\Delta_{b,n}/2} \\ &= \frac{8n|h_{ab}|^2 Q_n}{[N_b(P_d) + |h_{ab}|^2 Q_n] \Delta_{b,n}^2} + e^{-\Delta_{b,n}/2}. \end{aligned} \quad (40)$$

Since $Q_n = \Theta(n^{-1/2})$, the variance term is $\Theta(\sqrt{n})$. Hence

$$\frac{\Delta_{b,n}}{n^{1/4}} \rightarrow \infty \quad (41)$$

is sufficient to ensure

$$\mathbb{E}_{C_n}[P_e^{(n)}] \rightarrow 0. \quad (42)$$

Moreover,

$$\begin{aligned}
nI_{b,n} &= n \log \left(1 + \frac{|h_{ab}|^2 Q_n}{N_b(P_d)} \right) \\
&= \frac{|h_{ab}|^2}{N_b(P_d)} nQ_n + O(nQ_n^2) \\
&= (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + O(1).
\end{aligned} \tag{43}$$

Resolvability at Willie. We now control the discrepancy between the finite-codebook Gaussian mixture and the ensemble-induced Gaussian reference law; this will yield the complementary lower-side requirement on $\log M_n$. Let $p_{Z|X}(z|x)$ denote the conditional density of Willie's channel,

$$p_{Z|X}(z|x) = \frac{1}{\pi N_w(P_d)} \exp \left(-\frac{|z - g_{aw}x|^2}{N_w(P_d)} \right), \tag{44}$$

and let $\bar{p}_{1,n}(z)$ denote the single-letter density associated with the Gaussian reference distribution in (17),

$$\bar{p}_{1,n}(z) = \frac{1}{\pi[N_w(P_d) + |g_{aw}|^2 Q_n]} \exp \left(-\frac{|z|^2}{N_w(P_d) + |g_{aw}|^2 Q_n} \right). \tag{45}$$

By construction,

$$\bar{p}_{1,n}(z) = \mathbb{E}_X [p_{Z|X}(z|X)], \quad X \sim \mathcal{CN}(0, Q_n). \tag{46}$$

Define Willie's single-letter and n -letter information densities as

$$\imath_{w,n}(x; z) \triangleq \log \frac{p_{Z|X}(z|x)}{\bar{p}_{1,n}(z)}, \tag{47}$$

and

$$\imath_{w,n}^{(n)}(x^n; z^n) \triangleq \sum_{i=1}^n \imath_{w,n}(x_i; z_i). \tag{48}$$

The density of the reference distribution is therefore

$$\bar{p}_{1,n}^{(n)}(z^n) = \prod_{i=1}^n \bar{p}_{1,n}(z_i),$$

and the random codebook $\mathbf{C}_n$ induces

$$p_{1,\mathbf{C}_n}^{(n)}(z^n) = \frac{1}{M_n} \sum_{m=1}^{M_n} \prod_{i=1}^n p_{Z|X}(z_i | X_i(m)). \tag{49}$$

Hence,

$$\frac{p_{1,\mathbf{C}_n}^{(n)}(z^n)}{\bar{p}_{1,n}^{(n)}(z^n)} = \frac{1}{M_n} \sum_{m=1}^{M_n} \exp \left[\imath_{w,n}^{(n)}(X^n(m); z^n) \right]. \tag{50}$$

By symmetry of the i.i.d. random codebook, let $X^n(1)$ denote the transmitted codeword and generate Z^n according to $\prod_{i=1}^n p_{Z|X}(z_i | X_i(1))$. Then

$$\begin{aligned}
&\mathbb{E}_{\mathbf{C}_n} D \left(P_{1,\mathbf{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)} \right) \\
&= \mathbb{E} \log \left[\frac{1}{M_n} \sum_{m=1}^{M_n} \exp \left(\imath_{w,n}^{(n)}(X^n(m); Z^n) \right) \right].
\end{aligned} \tag{51}$$

For every $j \neq 1$, $X^n(j)$ is independent of Z^n , and

$$\begin{aligned}
\mathbb{E}_{X^n(j)} \exp \left[\imath_{w,n}^{(n)}(X^n(j); z^n) \right] &= \prod_{i=1}^n \frac{\mathbb{E}_X [p_{Z|X}(z_i | X_i)]}{\bar{p}_{1,n}(z_i)} \\
&= 1.
\end{aligned} \tag{52}$$

Conditioning on $(X^n(1), Z^n)$ and applying Jensen's inequality to $X^n(2), \dots, X^n(M_n)$ therefore gives

$$\begin{aligned}
\mathbb{E}_{\mathbf{C}_n} D \left(P_{1,\mathbf{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)} \right) &\leq \mathbb{E} \log \left[\frac{e^{\imath_{w,n}^{(n)}(X^n; Z^n)} + M_n - 1}{M_n} \right] \\
&= \mathbb{E} \log \left[1 + \frac{e^{\imath_{w,n}^{(n)}(X^n; Z^n)} - 1}{M_n} \right] \\
&\leq \mathbb{E} \log \left[1 + \frac{e^{\imath_{w,n}^{(n)}(X^n; Z^n)}}{M_n} \right].
\end{aligned} \tag{53}$$

For any fixed $0 < \alpha < 1$, using $\log(1+t) \leq t^\alpha/\alpha$ for $t \geq 0$ yields

$$\begin{aligned}
\mathbb{E}_{\mathbf{C}_n} D \left(P_{1,\mathbf{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)} \right) &\leq \frac{1}{\alpha M_n^\alpha} \mathbb{E} \exp \left[\alpha \imath_{w,n}^{(n)}(X^n; Z^n) \right] \\
&= \frac{1}{\alpha M_n^\alpha} \left(\mathbb{E} e^{\alpha \imath_{w,n}^{(n)}(X; Z)} \right)^n,
\end{aligned} \tag{54}$$

where the last equality follows from the memoryless channel and the i.i.d. Gaussian input.

Gaussian Exponential Moment at Willie. To make the preceding one-shot resolvability bound explicit, it remains to

evaluate the exponential moment of Willie's information density under the Gaussian input law. Define

$$s_{w,n} \triangleq \frac{|g_{aw}|^2 Q_n}{N_w(P_d)}. \quad (55)$$

Under the joint distribution of (X, Z) , define $U \triangleq \frac{g_{aw}X}{|g_{aw}|\sqrt{Q_n}}$, and $W \triangleq \frac{Z - g_{aw}X}{\sqrt{N_w(P_d)}}$. Then

$$U, W \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, 1), \quad \frac{Z}{\sqrt{N_w(P_d)}} = \sqrt{s_{w,n}}U + W.$$

Substitution into (42) gives

$$\iota_{w,n}(X; Z) = \log(1 + s_{w,n}) - |W|^2 + \frac{|\sqrt{s_{w,n}}U + W|^2}{1 + s_{w,n}}. \quad (56)$$

Let $\boldsymbol{\xi} \triangleq \begin{bmatrix} U \\ W \end{bmatrix} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_2)$. Then

$$\iota_{w,n}(X; Z) = \log(1 + s_{w,n}) + \boldsymbol{\xi}^H \mathbf{A}_{w,n} \boldsymbol{\xi}, \quad (57)$$

where

$$\mathbf{A}_{w,n} = \frac{1}{1 + s_{w,n}} \begin{bmatrix} s_{w,n} & \sqrt{s_{w,n}} \\ \sqrt{s_{w,n}} & -s_{w,n} \end{bmatrix}. \quad (58)$$

For $0 < \alpha < 1$, $\mathbf{I}_2 - \alpha \mathbf{A}_{w,n} \succ 0$, since $1 - \frac{\alpha s_{w,n}}{1 + s_{w,n}} > 0$ and

$$\det(\mathbf{I}_2 - \alpha \mathbf{A}_{w,n}) = 1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}} > 0. \quad (59)$$

Hence the proper-complex Gaussian integral gives

$$\begin{aligned} \mathbb{E} e^{\alpha \iota_{w,n}(X; Z)} &= (1 + s_{w,n})^\alpha \mathbb{E} e^{\alpha \boldsymbol{\xi}^H \mathbf{A}_{w,n} \boldsymbol{\xi}} \\ &= \frac{(1 + s_{w,n})^\alpha}{\det(\mathbf{I}_2 - \alpha \mathbf{A}_{w,n})} \\ &= \frac{(1 + s_{w,n})^\alpha}{1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}}}. \end{aligned} \quad (60)$$

Substituting into (49),

$$\mathbb{E}_{C_n} D\left(P_{1, C_n}^{(n)} \parallel \bar{P}_{1, n}^{(n)}\right) \leq \frac{1}{\alpha M_n^\alpha} \left[\frac{(1 + s_{w,n})^\alpha}{1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}}} \right]^n. \quad (61)$$

Asymptotic Resolvability. We now choose the codebook-size margin and an n -dependent moment parameter so that the explicit resolvability bound vanishes without affecting the first-order $\sqrt{n}$ scale. Define Willie's Gaussian-input mutual information as

$$I_{w,n} \triangleq \log(1 + s_{w,n}). \quad (62)$$

The preceding resolvability bound can be written, for any $0 < \alpha < 1$, as

$$\begin{aligned} \mathbb{E}_{C_n} D\left(P_{1, C_n}^{(n)} \parallel \bar{P}_{1, n}^{(n)}\right) &\leq \exp \left\{ -\log \alpha - \alpha (\log M_n - n I_{w,n}) - n \log \left(1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}} \right) \right\}. \end{aligned} \quad (63)$$

From the selected power sequence,

$$s_{w,n} = \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} = (1 - \varepsilon) \sqrt{\frac{2\delta}{n}}, \quad (64)$$

so that

$$s_{w,n} = \Theta(n^{-1/2}), \quad n s_{w,n} = \Theta(\sqrt{n}). \quad (65)$$

Consider codebook sizes satisfying

$$\log M_n - n I_{w,n} \geq n^{3/8}. \quad (66)$$

Since the one-shot bound holds for every $0 < \alpha < 1$ at each n , choose

$$\alpha_n \triangleq \frac{n^{3/8}}{4n s_{w,n}}. \quad (67)$$

Then

$$\alpha_n = \Theta(n^{-1/8}) \longrightarrow 0, \quad (68)$$

and hence $0 < \alpha_n < 1$ for all sufficiently large n . Moreover,

$$\frac{\alpha_n^2 s_{w,n}}{1 + s_{w,n}} \longrightarrow 0.$$

Using $-\log(1-u) \leq 2u$ for sufficiently small $u \geq 0$ gives

$$\begin{aligned} \mathbb{E}_{C_n} D\left(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) &\leq \exp\left\{-\log \alpha_n - \alpha_n n^{3/8} + \frac{2n\alpha_n^2 s_{w,n}}{1+s_{w,n}}\right\} \\ &\leq \exp\left\{-\log \alpha_n - \alpha_n n^{3/8} + 2n\alpha_n^2 s_{w,n}\right\} \\ &= \frac{4ns_{w,n}}{n^{3/8}} \exp\left(-\frac{n^{3/4}}{8ns_{w,n}}\right). \end{aligned} \quad (69)$$

Since $ns_{w,n} = \Theta(\sqrt{n})$,

$$\frac{4ns_{w,n}}{n^{3/8}} = O(n^{1/8}), \quad \frac{n^{3/4}}{8ns_{w,n}} = \Theta(n^{1/4}), \quad (70)$$

and therefore

$$\mathbb{E}_{C_n} D\left(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) \rightarrow 0. \quad (71)$$

Thus, a sufficient resolvability condition is

$$\log M_n \geq nI_{w,n} + n^{3/8}. \quad (72)$$

Thus, in contrast to reliability, resolvability requires the codebook to contain sufficiently many codewords to approximate the Gaussian reference output.

Compatibility and Codebook-Size Selection. Reliability imposes an upper requirement on the admissible codebook size, whereas resolvability imposes a lower requirement. It remains to verify that the two conditions admit a common interval and to select an integer M_n within it. The Gaussian-input mutual information at Willie satisfies

$$\begin{aligned} nI_{w,n} &= n \log \left(1 + \frac{|g_{aw}|^2 Q_n}{N_w(P_d)}\right) \\ &= \frac{|g_{aw}|^2}{N_w(P_d)} nQ_n + O(nQ_n^2) \\ &= (1-\varepsilon)\sqrt{2\delta}\sqrt{n} + O(1). \end{aligned} \quad (73)$$

Combining this with the corresponding expansion of $nI_{b,n}$ gives

$$n(I_{b,n} - I_{w,n}) = (1-\varepsilon)\sqrt{2\delta} \left[\frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} - 1 \right] \sqrt{n} + O(1). \quad (74)$$

Assume

$$\frac{|h_{ab}|^2}{N_b(P_d)} > \frac{|g_{aw}|^2}{N_w(P_d)}. \quad (75)$$

Then

$$n(I_{b,n} - I_{w,n}) = \Theta(\sqrt{n}), \quad (76)$$

with a strictly positive leading coefficient. Since $n^{3/8} = o(\sqrt{n})$, there exists n_0 such that, for all $n \geq n_0$,

$$n(I_{b,n} - I_{w,n}) \geq 2n^{3/8} + \log 2. \quad (77)$$

For such n , define

$$M_n \triangleq \left\lfloor \exp\left(nI_{b,n} - n^{3/8}\right) \right\rfloor. \quad (78)$$

Since $nI_{b,n} - n^{3/8} \rightarrow \infty$, for all sufficiently large n ,

$$nI_{b,n} - n^{3/8} - \log 2 \leq \log M_n \leq nI_{b,n} - n^{3/8}. \quad (79)$$

Consequently,

$$nI_{b,n} - \log M_n \geq n^{3/8}, \quad (80)$$

and

$$\begin{aligned} \log M_n - nI_{w,n} &\geq n(I_{b,n} - I_{w,n}) - n^{3/8} - \log 2 \\ &\geq n^{3/8}. \end{aligned} \quad (81)$$

Hence the selected M_n simultaneously satisfies the sufficient conditions for Bob reliability and Willie resolvability.

Moreover,

$$\begin{aligned} \log M_n &= nI_{b,n} - n^{3/8} + O(1) \\ &= (1-\varepsilon)\sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + o(\sqrt{n}). \end{aligned} \quad (82)$$

Per-Codeword Power Constraint. Gaussian codeword generation satisfies the power constraint only probabilistically, so we must verify that all M_n codewords obey the prescribed per-codeword bound simultaneously. Define the event that all codewords in the random codebook satisfy the prescribed power constraint as

$$\mathcal{G}_n \triangleq \left\{ \frac{1}{n} \|X^n(m)\|^2 \leq P_a, \forall m \in \{1, \dots, M_n\} \right\}. \quad (83)$$

Its complement is therefore

$$\mathcal{G}_n^c = \left\{ \exists m \in \{1, \dots, M_n\} : \frac{1}{n} \|X^n(m)\|^2 > P_a \right\}. \quad (84)$$

Since $P_a > 0$ is fixed and $Q_n = \Theta(n^{-1/2})$,

$$\frac{P_a}{Q_n} \rightarrow \infty. \quad (85)$$

For any fixed codeword index m , the Gaussian construction gives

$$\frac{|X_i(m)|^2}{Q_n} \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(1), \quad i = 1, \dots, n,$$

and hence, for every $0 < t < 1$,

$$\mathbb{E} \exp\left(t \frac{|X_i(m)|^2}{Q_n}\right) = \frac{1}{1-t}.$$

Applying the Chernoff bound,

$$\begin{aligned} \Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] &= \Pr\left[\sum_{i=1}^n \frac{|X_i(m)|^2}{Q_n} > \frac{nP_a}{Q_n}\right] \\ &\leq \exp\left(-\frac{tnP_a}{Q_n}\right) \prod_{i=1}^n \mathbb{E} \exp\left(t \frac{|X_i(m)|^2}{Q_n}\right) \\ &= \exp\left\{-n \left[\frac{tP_a}{Q_n} + \log(1-t)\right]\right\}. \end{aligned} \quad (86)$$

Choosing $t = 1/2$ yields

$$\Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] \leq \exp\left\{-n \left[\frac{P_a}{2Q_n} - \log 2\right]\right\}. \quad (87)$$

Since $P_a/Q_n \rightarrow \infty$, for all sufficiently large n ,

$$\frac{P_a}{4Q_n} \geq \log 2,$$

and therefore

$$\Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] \leq \exp\left(-\frac{nP_a}{4Q_n}\right). \quad (88)$$

Using the definition of $\mathcal{G}_n^c$ and a union bound over all codewords,

$$\begin{aligned} \Pr(\mathcal{G}_n^c) &= \Pr\left[\bigcup_{m=1}^{M_n} \left\{\frac{1}{n}\|X^n(m)\|^2 > P_a\right\}\right] \\ &\leq \sum_{m=1}^{M_n} \Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] \\ &= M_n \Pr\left[\frac{1}{n}\|X^n(1)\|^2 > P_a\right] \\ &\leq \exp\left\{\log M_n - \frac{nP_a}{4Q_n}\right\}. \end{aligned} \quad (89)$$

For the previously selected codebook size,

$$\log M_n = O(\sqrt{n}), \quad (90)$$

whereas

$$Q_n = \Theta(n^{-1/2}) \quad \implies \quad \frac{nP_a}{Q_n} = \Theta(n^{3/2}). \quad (91)$$

Consequently,

$$\log M_n - \frac{nP_a}{4Q_n} = -\Theta(n^{3/2}), \quad (92)$$

and hence

$$\Pr(\mathcal{G}_n^c) \longrightarrow 0, \quad \Pr(\mathcal{G}_n) \longrightarrow 1. \quad (93)$$

Thus, with probability tending to one under the random-codebook ensemble, all codewords simultaneously satisfy

$$\frac{1}{n}\|X^n(m)\|^2 \leq P_a, \quad m = 1, \dots, M_n. \quad (94)$$

Actual Coverttness. Resolvability controls the distance to the Gaussian reference law rather than directly to the innocent distribution. We therefore connect the two through the exact ensemble-average output identity. Let $p_0^{(n)}(z^n)$ denote the density of $P_0^{(n)}$. We first verify that the Gaussian reference distribution is exactly the ensemble-average output distribution induced by the random codebook. For every $z^n \in \mathbb{C}^n$,

$$\begin{aligned} \mathbb{E}_{\mathcal{C}_n} \left[p_{1, \mathcal{C}_n}^{(n)}(z^n) \right] &= \frac{1}{M_n} \sum_{m=1}^{M_n} \mathbb{E}_{X^n(m)} \left[\prod_{i=1}^n p_{Z|X}(z_i | X_i(m)) \right] \\ &= \prod_{i=1}^n \mathbb{E}_X \left[p_{Z|X}(z_i | X) \right] = \prod_{i=1}^n \bar{p}_{1,n}(z_i) = \bar{p}_{1,n}^{(n)}(z^n). \end{aligned} \quad (95)$$

Hence,

$$\mathbb{E}_{\mathcal{C}_n} \left[P_{1, \mathcal{C}_n}^{(n)} \right] = \bar{P}_{1,n}^{(n)}. \quad (96)$$

For every realization of the random codebook,

$$\begin{aligned} D\left(P_{1, \mathcal{C}_n}^{(n)} \parallel P_0^{(n)}\right) &= D\left(P_{1, \mathcal{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) \\ &\quad + \int_{\mathbb{C}^n} p_{1, \mathcal{C}_n}^{(n)}(z^n) \log \frac{\bar{p}_{1,n}^{(n)}(z^n)}{p_0^{(n)}(z^n)} dz^n. \end{aligned} \quad (97)$$

Taking expectation over $\mathbf{C}_n$ and using the preceding ensemble-average identity gives

$$\begin{aligned} \mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel P_0^{(n)}\right) \\ = \mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) + D\left(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}\right). \end{aligned} \quad (98)$$

Since

$$s_{w,n} = (1 - \varepsilon) \sqrt{\frac{2\delta}{n}},$$

the reference divergence is

$$D\left(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}\right) = n[s_{w,n} - \log(1 + s_{w,n})]. \quad (99)$$

For every $x \geq 0$,

$$x - \log(1 + x) \leq \frac{x^2}{2}. \quad (100)$$

Indeed,

$$\frac{d}{dx} \left[\frac{x^2}{2} - x + \log(1 + x) \right] = \frac{x^2}{1 + x} \geq 0,$$

and the expression in brackets equals zero at $x = 0$. Therefore,

$$\begin{aligned} D\left(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}\right) &\leq \frac{ns_{w,n}^2}{2} \\ &= (1 - \varepsilon)^2 \delta. \end{aligned} \quad (101)$$

Moreover, the preceding resolvability result gives

$$\mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) \rightarrow 0. \quad (102)$$

Since

$$\delta - (1 - \varepsilon)^2 \delta = \delta(2\varepsilon - \varepsilon^2) > 0, \quad (103)$$

there exists n_0 such that, for every $n \geq n_0$,

$$\mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) \leq \frac{\delta(2\varepsilon - \varepsilon^2)}{2}. \quad (104)$$

Consequently,

$$\begin{aligned} \mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel P_0^{(n)}\right) &\leq (1 - \varepsilon)^2 \delta + \frac{\delta(2\varepsilon - \varepsilon^2)}{2} \\ &= \delta - \frac{\delta(2\varepsilon - \varepsilon^2)}{2} \\ &< \delta. \end{aligned} \quad (105)$$

Finally, since KL divergence is nonnegative, Markov's inequality gives, for all sufficiently large n ,

$$\begin{aligned} \Pr\left[D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel P_0^{(n)}\right) > \delta\right] \\ \leq \frac{\mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel P_0^{(n)}\right)}{\delta} \\ \leq 1 - \frac{2\varepsilon - \varepsilon^2}{2}. \end{aligned} \quad (106)$$

Hence,

$$\Pr\left[D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel P_0^{(n)}\right) \leq \delta\right] \geq \frac{2\varepsilon - \varepsilon^2}{2} > 0. \quad (107)$$

Thus, for all sufficiently large n , the random-codebook ensemble contains codebook realizations satisfying the actual covertness constraint with probability bounded away from zero.

Deterministic Fixed-Codebook Extraction. The preceding arguments establish the required properties at the ensemble level or with high probability. We finally show that a single deterministic realization satisfies reliability, actual covertness, and the power constraint simultaneously. For a realization $\mathbf{C}_n$ of the random codebook, let $P_e^{(n)}(\mathbf{C}_n)$ denote the average decoding error probability of its associated ML decoder. The preceding reliability analysis gives

$$\mathbb{E}_{\mathbf{C}_n} \left[P_e^{(n)}(\mathbf{C}_n) \right] \rightarrow 0. \quad (108)$$

Since $P_e^{(n)}(\mathbf{C}_n) \geq 0$, Markov's inequality yields

$$\begin{aligned} \Pr\left[P_e^{(n)}(\mathbf{C}_n) > \sqrt{\mathbb{E}_{\mathbf{C}_n} \left[P_e^{(n)}(\mathbf{C}_n) \right]}\right] \\ \leq \sqrt{\mathbb{E}_{\mathbf{C}_n} \left[P_e^{(n)}(\mathbf{C}_n) \right]} \rightarrow 0. \end{aligned} \quad (109)$$

The power-constraint analysis has established

$$\Pr(\mathcal{G}_n^c) \rightarrow 0, \quad (110)$$

where $\mathcal{G}_n$ is the event that every codeword satisfies the per-codeword power constraint. Moreover, the actual-covertness analysis gives, for all sufficiently large n ,

$$\Pr\left[D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel P_0^{(n)}\right) \leq \delta\right] \geq \frac{2\varepsilon - \varepsilon^2}{2} > 0. \quad (111)$$

Using the union bound on their complements,

$$\begin{aligned}
& \Pr \left[\mathcal{G}_n, P_e^{(n)}(\mathbf{C}_n) \leq \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]}, \right. \\
& \quad \left. D(P_{1,\mathbf{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta \right] \\
& \geq \Pr \left[D(P_{1,\mathbf{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta \right] \\
& \quad - \Pr(\mathcal{G}_n^c) \\
& \quad - \Pr \left[P_e^{(n)}(\mathbf{C}_n) > \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]} \right] \\
& \geq \frac{2\varepsilon - \varepsilon^2}{2} - \Pr(\mathcal{G}_n^c) - \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]}.
\end{aligned} \tag{112}$$

The last two terms vanish as $n \rightarrow \infty$, whereas $(2\varepsilon - \varepsilon^2)/2 > 0$. Hence, for all sufficiently large n ,

$$\Pr \left[\mathcal{G}_n, P_e^{(n)}(\mathbf{C}_n) \leq \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]}, D(P_{1,\mathbf{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta \right] > 0. \tag{113}$$

Therefore, for every sufficiently large n , there exists a deterministic realization

$$\mathcal{C}_n^* \tag{114}$$

of the random codebook such that

$$P_e^{(n)}(\mathcal{C}_n^*) \leq \sqrt{\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}(\mathbf{C}_n)]} \rightarrow 0, \tag{115}$$

$$D(P_{1,\mathcal{C}_n^*}^{(n)} \| P_0^{(n)}) \leq \delta, \tag{116}$$

and

$$\frac{1}{n} \|x^n(m)\|^2 \leq P_a, \quad m = 1, \dots, M_n. \tag{117}$$

Thus $\{\mathcal{C}_n^*\}$ forms a sequence of deterministic fixed codebooks satisfying reliability, actual covertness, and the per-codeword power constraint simultaneously.

Since every realization of $\mathbf{C}_n$ contains exactly M_n codewords, the extraction does not alter the previously selected codebook size. Hence

$$\log M_n = (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + o(\sqrt{n}). \tag{118}$$

Achievable Square-Root Coefficient. Since deterministic extraction preserves the selected codebook size, it remains only to identify the resulting first-order coefficient and remove the arbitrary backoff through the supremum over ε . For every fixed $0 < \varepsilon < 1$, the preceding construction yields a sequence of deterministic fixed codebooks $\{\mathcal{C}_n^*\}$ satisfying, for all sufficiently large n ,

$$P_e^{(n)}(\mathcal{C}_n^*) \rightarrow 0, \tag{119}$$

$$D(P_{1,\mathcal{C}_n^*}^{(n)} \| P_0^{(n)}) \leq \delta, \tag{120}$$

and

$$\frac{1}{n} \|x^n(m)\|^2 \leq P_a, \quad m = 1, \dots, M_n. \tag{121}$$

Moreover, deterministic-codebook extraction does not alter the previously selected codebook size, and hence

$$\begin{aligned}
\log M_n &= nI_{b,n} - n^{3/8} + O(1) \\
&= (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + o(\sqrt{n}).
\end{aligned} \tag{122}$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{\log M_n}{\sqrt{n}} = (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}. \tag{123}$$

Since $\mathcal{C}_n^*$ is an admissible deterministic codebook, the optimal codebook size satisfies

$$M_n^*(\delta) \geq M_n \tag{124}$$

for all sufficiently large n . Consequently,

$$\liminf_{n \rightarrow \infty} \frac{\log M_n^*(\delta)}{\sqrt{n}} \geq (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}. \tag{125}$$

The above inequality holds for every fixed $0 < \varepsilon < 1$. Taking the supremum over ε and using

$$\sup_{0 < \varepsilon < 1} (1 - \varepsilon) = 1$$

gives

$$\boxed{\liminf_{n \rightarrow \infty} \frac{\log M_n^*(\delta)}{\sqrt{n}} \geq \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}}. \tag{126}$$

□